Midterm Review

6 Questions in midterm exam

Q1: 8 subquestions (25 points)

Q2 - Q6: computation. proof. 2 hrs

3 2

1st week.

Computation error : truncation error , round-off error

$f(x) = P_n(x) + R(x)$

$\frac{1}{3} = 0.3333\cdots3(\infty)$

unavoidable

absolute error $|p - p^*|$

relative error $\frac{|p - p^*|}{|p|}$

Significant digits:

$$\frac{|p - p^*|}{|p|} \leq \frac{1}{2} \times 10^{-t}$$

chopping Vs. rounding (四舍五入)

Avoid loss of accuracy due to round-off error

Example: $x = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$ $\quad b = 6.01$

$a = c = 1$

Reformulate $x = \dfrac{b^2 - b^2 + 4ac}{2a\left(-b - \sqrt{b^2 - 4ac}\right)}$

$= \dfrac{-2c}{b + \sqrt{b^2 - 4ac}}$

$\left(\dfrac{x}{y}\right)$ when $y << 1$

$$\frac{x + \varepsilon_x}{y + \varepsilon_y} = \frac{x}{y + \varepsilon_y} + \frac{\varepsilon_x}{y + \varepsilon_y}$$

Rewrite polynomials into nested form ?

More efficient and reduce round-off error

Ex. $f(x) = x^3 - 6.1x^2 + 3.2x + 1.5$

$= ((x - 6.1)x + 3.2)x + 1.5$

Avoid large numbers eat small numbers.

2nd week.

1. Bisection method
   1. Existence
   2. the algorithm. (Shaving the interval)
   3. error analysis

$$|p_n - p| \le \frac{b-a}{2^n}$$

$|p_n - p|$: analysis the error

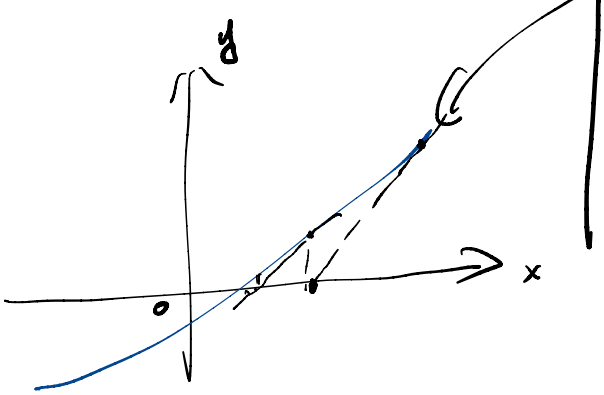
$\frac{b-a}{2^n}$: Find some N to achieve some accuracy

2. Fixed-point iteration
   1. Existence of the fixed point
   2. algorithm, $x^{n+1} = g(x^n)$
   3. uniqueness ($|g'(x)| \le k < 1$)
   4. convergence analysis
   5. convergence rate

$$|p_n - p| \le \frac{k^n}{1-k} |p_1 - p_0|$$

3rd + 4th week.

Newton's method:



1. 2nd order method
2. Geometric interpretation
3. Convergence proof
4. Convergence rate

Secant method ← $\frac{1+\sqrt{5}}{2}$

↑

the proof

the algorithm

Müller's method ← 1.84

False position method

5th Week.

Interpolation: Lagrange: $L_2 = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x-x_1)}$

$L_0 =$ —

$L_1 =$ — ✓

Error analysis + proof ✓.

Runge's phenomenon ?

Neville's method

Q_{00}			
Q_{10}	Q_{11}		
Q_{20}	Q_{21}	Q_{22}	

Divided difference $f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

Theorem on Page 9 of Week 5-2.

6th week:

Hermite polynomial; Divided difference

7th week

Spline: nature spline V.S. clamped spline?

Construction?

8th week

Numerical differentiation:

Forward / backward difference } 1st order derivation

three-point - - - -

three-point formula → 2nd order

Truncation error analysis

Richardson extrapolation.